

Part B-1

1. Excellence

1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

MERISTIME — From thermosensing to organ timing in the shoot apical meristem

Introduction.

Unlike animals, whose body plan is essentially fixed during embryogenesis, plants build most of their body after germination. This post-embryonic mode of development is the source of their remarkable plasticity, and it depends on the sustained activity of meristems, a specific tissue containing stem cells that give rise to all plant organs.

The shoot apical meristem (SAM) originates all aerial organs leading to the initiation of leaves, stems and flowers; during vegetative development, its activity sets the pace for leaf initiation rate, known as the **plastochron**. Temperature is a central regulator of this rate: within a biologic range, warmer temperature accelerates leaf production, and this acceleration is so regular that the rate of leaf initiation is linearly related to thermal time — the number of degree-days the plant experiences over a given period. This relationship is conserved from *Arabidopsis* to crops and is a major determinant of development and yield. While temperature signaling has been dissected in detail in the context of hypocotyl elongation, its action within the SAM, where the organs that shape the plant are born, **remains almost completely unexplored**. This project aims to decipher how the SAM perceives and transduces temperature cues to set the plastochron and proposes that a missing piece might be the redox and nitric oxide (NO) signaling.

State of the art and objectives.

The SAM is organized into functional zones. In the central zone, a WUSCHEL (WUS)–CLAVATA3 (CLV3) feedback loop maintains stem cell identity; in the peripheral zone, faster-dividing cells differentiate into leaf primordia at sites specified by PIN1-dependent auxin maxima, with cytokinin promoting leaf proliferation. Recent 3D imaging showed that temperature acts directly within the SAM, altering meristem size and WUS/CLV3 expression patterns¹. However, **how these changes drive leaf initiation, whether temperature signals through auxin or cytokinin pathways, and how temperature perception is first transduced into SAM activity remain unknown**.

Temperature-sensing has been described in the hypocotyl: phytochrome B (phyB) acts as a thermosensor inactivated at warm temperature⁴; ELF3, a transcriptional regulator, also gets inactivated under warm temperature; PIF7 whose transcription increases through temperature-dependent mRNA translation; and warm temperature drives starch breakdown and sucrose production. Downstream, signalling converges on PIF4 and the E3 ligase COP1⁵. Whether these mechanisms also operate in the SAM to control leaf initiation is unknown.

In parallel, it is known that redox species constitutes key signals within the SAM. The meristem is also redox-compartmentalised: superoxide accumulates in the central zone, sustaining WUS and stemness, whereas hydrogen peroxide and NO are enriched in the differentiating peripheral zone, and disrupting this gradient reduces SAM size and delays early leaf development^{2–3}. The peripheral zone is precisely where primordia are initiated, so the NO-rich domain coincides with the site where the plastochron is literally set. NO promotes peripheral-zone fate by repressing WUS, in part by limiting the stem-cell regulator AGO4 to the centre through S-nitrosylation, with WUS interacting with AGO4 in a NO-dependent manner³.

Preliminary results.

Using a thermal-time phenotyping assay restricted to the vegetative phase, I have compared wild type and the thermosensor mutant phyB across temperatures. Wild type holds the plastochron on an essentially constant

thermal-time relationship, whereas phyB does not: its leaf-initiation rate departs from the thermal-time expectation (Fig. 1). This identifies a meristem-level temperature phenotype for phyB, distinct from its established role in hypocotyl elongation, and it is the observation on which the project is built. I have also begun to manipulate NO in the apex with the donor GSNO and the scavenger cPTIO, read on the same thermal-time basis; these experiments are in progress and have not yet produced a clear plastochron phenotype, and establishing whether NO acts downstream of temperature in the meristem is the central question of WP2. The reporter lines (pCLV3, pWUS, DR5, PIN1) and the CherryTemp-based live-imaging required for WP1 are in place in the host laboratory.

[Figure 1 — to be supplied by the applicant: plastochron vs thermal time, wild type vs phyB.]

Hypothesis and objectives. My hypothesis is that temperature is transduced in the SAM through a redox/NO node that remodels the peripheral-zone redox landscape to set the plastochron. **The general aim is to understand the early mechanisms linking temperature perception to plastochron regulation in the SAM.**

WP1: Characterisation of the temperature response of the meristem during early development. I aim to describe the effect of temperature on SAM function and leaf initiation with high spatial and temporal resolution (plastochron on thermal-time basis, SAM size, WUS/CLV3 expression and distribution).

WP2: Role of phyB and of redox/NO signalling in the temperature control of leaf initiation. I will test whether the canonical thermosensor phyB and the meristem redox/NO system transduce temperature into plastochron regulation. Task 2.1 addresses the general redox state of the meristem — the superoxide/H₂O₂ balance across the central and peripheral zones — as a function of temperature; Task 2.2 addresses nitric oxide specifically, through in vivo detection, NO donors and scavengers, and NO-pathway mutants, asking whether NO perturbation reproduces the phyB phenotype established in my preliminary work.

WP3: Single-cell transcriptomic response of the SAM to warm temperature. Using a factorial temperature × thermal-time design, I will characterise the meristem's temperature response at cellular resolution and test whether known meristem regulators — the WUS/CLV3 network, zonal redox and NO-biosynthesis genes, and auxin readouts — participate in temperature and thermal-time sensing. This yields the temperature- and redox-responsive gene networks of the SAM and a reusable community dataset, and is carried out with R. Reis (University of Bern).

Originality and ambition. Three features make the project original. First, it moves temperature signalling out of the hypocotyl, where it is almost exclusively studied, and into the shoot apical meristem, where the organs that shape the plant are initiated. Second, it proposes a specific, testable transducer — a redox/NO node in the peripheral zone — rather than treating meristem thermosensing as a generic unknown: my preliminary data establish that phyB is required to hold the plastochron on a constant thermal-time relationship, a meristem phenotype distinct from the canonical hypocotyl-elongation output, and I propose that phyB acts here through a redox/NO route rather than the PIF4/COP1 growth-effector branch. Third, it connects two fields that have developed in parallel — thermomorphogenesis, which has identified the sensors, and the redox biology of the stem-cell niche, which provides the positional signals that pattern organ initiation. Establishing how an environmental cue is converted into the timing of organ formation is of broad significance for developmental biology and directly relevant to crop resilience under a warming climate, where the rate of leaf initiation shapes canopy establishment and yield.

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices)

Overall approach. The methodology pairs two logics rarely combined in one laboratory: the quantitative, imaging-based developmental biology of the shoot apical meristem, and the biochemistry of nitric oxide and redox signalling. A single output — the rate and pattern of organ initiation, expressed on a thermal-time basis — is read through three complementary lenses (genetic and phenotypic, in vivo imaging, and biochemical), so that no central claim rests on one technique. All experiments are restricted to the vegetative phase and controlled for flowering time, and thermal time is computed from logged growth-chamber temperatures, so that chronological age and developmental progression are separated throughout.

WP1 — Characterisation of the temperature response of the meristem during early development.

Task 1.1 — Plastochron and phyllotaxis against thermal time. I will quantify the plastochron and the phyllotactic divergence angle in wild type and selected mutants (phyB and the NO-pathway mutants used in WP2), grown at contrasting constant temperatures and under defined temperature shifts. Per-genotype linear models of leaf number against thermal time are compared statistically to establish whether each genotype maintains a constant thermal-time relationship (compensation) or departs from it.

Task 1.2 — Live imaging of meristem and auxin reporters under controlled temperature. Using the stem-cell reporters pCLV3 and pWUS and the auxin reporters DR5 and PIN1, I will image the living apex by confocal microscopy on the spectral confocal microscope recently acquired by the Institute of Biology (SNSF R'equip, 2024), with temperature controlled at the sample by the CherryTemp system available in the host laboratory (stable set-points and rapid shifts during time-lapse). Readouts — SAM size, the size and position of the WUS and CLV3 domains, the position of PIN1-dependent auxin maxima, and the timing of primordium emergence — are each placed on a thermal-time axis.

WP2 — Role of phyB and of redox/NO signalling in the temperature control of leaf initiation.

Task 2.1 — Temperature and the redox compartmentalisation of the meristem. I will map reactive redox species in the living apex across temperatures and in phyB: the superoxide/hydrogen peroxide balance (DHE and NBT for $O_2^{\cdot-}$; H_2DCFDA and DAB for H_2O_2), quantified by confocal imaging and histochemistry. The question is whether warm temperature reconfigures the central-versus-peripheral redox pattern described by Zeng et al.²⁻³, and whether any such change depends on phyB.

Task 2.2 — Nitric oxide as a candidate temperature transducer. NO is detected in vivo with DAF-FM DA and perturbed genetically (gsnor1/hot5, nia1 nia2, noa1) and pharmacologically (the donor GSNO and the scavenger cPTIO; glutathione as a control for the general thiol-redox context, not as an NO-specific treatment). All perturbations are read on a thermal-time basis and compared directly with the phyB phenotype. My preliminary data establish that phyB fails to hold the plastochron on a constant thermal-time relationship; NO manipulations are in progress and have not yet produced a clear phenotype. The decisive experiment of this task is whether NO perturbation reproduces this loss of thermal-time compensation. NO-pathway mutants are crossed with the DR5, PIN1, WUS and CLV3 reporters to localise any effect within the meristem, and candidate S-nitrosylation targets in the stem-cell niche (e.g. AGO4) are probed by biotin-switch assays.

WP3 — Single-cell transcriptomic response of the SAM to warm temperature.

Comparing two temperatures at equal chronological age confounds temperature with developmental stage (warm develops faster); comparing at equal thermal time controls development but not chronological age.

Task 3.1 — Factorial temperature × thermal-time design. I will profile dissected vegetative apices by single-nucleus RNA-seq under a factorial design: two temperatures (control vs warm) × two matched thermal-time points, rather than a gradient, which for single-cell work would dilute replication and power. This separates the effect of temperature at matched development from the effect of developmental progression and provides a direct test of compensation: a gene genuinely compensated on a thermal-time basis should not differ between temperatures once thermal time is matched.

Task 3.2 — Analysis and candidate networks. Predictions from Zeng et al.²⁻³ anchor quality control and a targeted analysis panel: WUS/CLV3 targets, AGO4 and RdDM components, zonal ROS-metabolising enzymes, the NO-biosynthesis genes NIA1, NIA2 and NOA1, and auxin readouts. Library preparation and single-cell analysis are carried out in collaboration with R. Reis (University of Bern), who provides the single-cell transcriptomics expertise; short technical visits to Bern are foreseen for the wet-lab and computational steps.

Interdisciplinarity. The project sits at the interface of developmental biology, plant physiology and redox biochemistry. Its feasibility rests on combining expertise that does not currently coexist in one place: the host's developmental and photobiological framework and quantitative live-imaging, my own redox/NO biochemistry and in vivo detection, and the single-cell transcriptomics of the Reis group.

Open science. Data are managed under a FAIR-compliant Data Management Plan prepared in the first months and updated thereafter. Datasets underlying publications — imaging, quantitative phenotyping and the single-

nucleus dataset — are deposited in recognised community repositories with persistent identifiers. Publications are made immediately open access and deposited in Libra, the institutional repository of the University of Neuchâtel, with preprints posted at submission. Protocols and analysis code, including the thermal-time image-analysis and statistical pipelines, are shared openly, and confirmatory analyses are pre-registered where possible.

Sex and gender dimension. This project is performed in the plant *Arabidopsis thaliana*, so a sex and gender dimension does not apply to its research methodology; however, gender dimension is considered.

1.3 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

What I gain from the host. My doctoral training in Argentina was in NO and redox signalling in plant physiology, at the level of the whole plant and of biochemical readouts. This fellowship provides what I do not yet have: the capacity to ask developmental questions at cellular resolution and in real time. This research group works at the interface of light and temperature signalling and shoot development, making it the environment where I can convert my signalling expertise into developmental biology. My training plan centres on three scientific competences the project requires and that I do not yet have: quantitative live-imaging of the meristem (reporter lines, temperature-controlled imaging with CherryTemp, 3D image-analysis pipelines); the genetics and conceptual framework of thermomorphogenesis, a field to which my supervisor is a foundational contributor; and single-nucleus transcriptomics together with its computational analysis. Technical training is provided within the host group and, for the single-cell work, through the collaboration with R. Reis (Bern) and dedicated courses of the Swiss Institute of Bioinformatics (bulk and single-cell RNA-seq analysis). Transferable-skills and career training is provided by the UniNE Graduate Campus (part of the swissuniversities 2025–2028 programme for the promotion of academic talent) and by the doctoral programmes of UniNE and the CUSO; I will also attend the EMBO Laboratory Leadership for Postdocs course and present at international plant photobiology and thermomorphogenesis meetings, including those co-organised by my supervisor in 2026.

What I bring to the host. The host group has no in vivo redox capability. I bring the detection of reactive redox species in living tissue (DAF-FM DA for NO; H₂DCFDA and DHE for ROS), the genetics of NO homeostasis, the pharmacological manipulation of NO with donors and scavengers, and the biochemistry of cysteine-based post-translational modifications, including biotin-switch workflows. The group's interest in redox as an integrator of environmental and endogenous cues in the meristem is explicit and current — for example in the shade-induced ROS/NO work my supervisor co-authored⁷ — but the technical capacity to pursue it in vivo is not yet established in Neuchâtel. My arrival makes that line executable and installs a capability that will outlive the fellowship: a genuine two-way transfer of knowledge.

Quality of supervision. Prof. Martina Legris leads a research group at the Institute of Biology, University of Neuchâtel. She specialises in the molecular signalling of photoreceptors and in developmental plasticity in response to light and temperature, with over 20 publications including first- or corresponding-author papers in *Science*, *PNAS*, *Nature Communications*, *New Phytologist* and *Plant Physiology*, and she received the 2023 *New Phytologist* Tansley Medal. Her contribution is foundational: she is first author of the work establishing phyB as a thermosensor that integrates light and temperature in *Arabidopsis*⁴, and she has since developed the study of light and temperature control of leaf and meristem morphogenesis^{5–7}. She trained in the Casal and Fankhauser laboratories and is herself a former MSCA, EMBO and HFSP fellow, so she knows the fellowship from the inside. My project is a direct extension of a question her own work opened — how temperature is read in the meristem to set leaf initiation — which makes the supervision scientifically substantive.

The group comprises the supervisor, one postdoctoral researcher and one PhD student, with a second postdoc (myself) joining on this project; the supervisor has additionally mentored numerous master's and summer students. It is funded by an SNSF Starting Grant (TMSGI3_218178, "Environmental control of shoot architecture in *Arabidopsis*", 2023–2028) and by a two-year Fondation Mercier pour la Science grant ("Temperature regulation of leaf initiation in *Arabidopsis thaliana*", 2026) that establishes the research line this fellowship contributes to, with access to a spectral confocal microscope co-funded by the supervisor (SNSF R'équip, 2024). Supervision is organised around weekly one-to-one and group meetings, a Career

Development Plan agreed in the first months and reviewed at regular intervals, and quarterly progress presentations. In the second year I will co-supervise, with the supervisor's support, a master's student on a defined part of the project.

Secondment. No secondment is foreseen. The single-nucleus transcriptomics of WP3 is carried out through a scientific collaboration with the group of Dr R. Reis (University of Bern), involving short technical visits rather than a formal secondment.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

I obtained my PhD in Exact Sciences (Biological Sciences and Biotechnology) with special distinction from the National University of La Plata in 2025, at the Institute of Plant Physiology (INFIVE, UNLP-CONICET), on the role of NO in acclimation to phosphorus restriction. I established that nitrate reductase mediates NO generation and that this NO is required for acclimation responses, published as first author⁸, and contributed to five further papers including reviews that helped define the agenda linking NO signalling to nutrient use efficiency. My work received the Best Poster Prize at the XXXV Meeting of the Argentine Society of Plant Physiology (2025) and I was awarded a competitive travel grant to present the flash talk "At the apex of thermomorphogenesis" at the 2026 thermomorphogenesis meeting in Dundee, which reflects the reception of this new line of work by the very community this project addresses.

I bring to the lab first, technical skills built by necessity: I established and applied *in vivo* redox imaging, enzymatic and HPLC determinations, biochemical analysis and hydroponic physiology, in a setting where techniques are more often built than bought. Second, early independence: I supervised an undergraduate project student for three years, taught as a teaching assistant, and sought out collaborations of my own, including with Lamattina's group in Mar del Plata (2021), which resulted in a co-authored paper⁹, and a research stay in the Chromosome Dynamics group at IPS2, Université Paris-Saclay (2025), whose expertise in chromatin biology and DNA methylation connects directly to the AGO4/RdDM dimension of the redox–meristem link addressed in WP2 and WP3. Third, sustained science outreach, which I will continue in Neuchâtel.

My doctorate gave me a clear identity as a redox biologist, and this fellowship is the moment to put that identity to work on questions I could not ask alone. Moving into developmental biology is a deliberate choice: redox signalling is already recognised as a regulator of meristem activity, but the tools to follow it *in vivo* and to connect it to temperature are not established there, and that is what I bring. Throughout my training in Argentina and in France I learned that science moves through people, and that open, collaborative practice is what makes knowledge grow. That is why I share materials and protocols openly and invest time in outreach, and why I intend to publish this project's results in open access and to deposit its data and protocols in public repositories. My long-term objective is to lead an independent group on how environmental signals are transduced by redox chemistry into developmental decisions.

2. Impact

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to their skills development

My long-term objective is to lead an independent research group at the interface of redox biology and plant development. This fellowship is the decisive step towards that goal: it adds to my established expertise in nitric oxide and redox physiology the three competences this research programme requires and that I currently lack — quantitative live-imaging of development at cellular resolution, the genetics and conceptual framework of thermomorphogenesis, and single-cell transcriptomics with its computational analysis. Acquiring these in a group working at the interface of light, temperature and shoot development positions me to lead a distinctive line of research that few others can, precisely because it joins two communities that rarely meet.

My skills development is organised around a Career Development Plan, agreed with my supervisor in the first months of the fellowship and reviewed at regular intervals, which sets objectives for research training, transferable skills and career progression. Scientific training follows the work packages: WP1 and WP2 build

my competence in reporter-based confocal imaging, temperature-controlled live-imaging and image quantification, while WP3, with the Reis group in Bern, introduces me to single-nucleus transcriptomics and the bioinformatic analysis of complex datasets, an increasingly indispensable skill in plant developmental biology.

Transferable skills are supported by the host institution. The UniNE Graduate Campus, established under the swissuniversities 2025–2028 programme for the promotion of academic talent, offers workshops and courses for the career development of postdoctoral researchers, and I will also have access, subject to availability, to courses of the doctoral programmes of UniNE and the CUSO. I will use this provision to strengthen the competences an independent researcher needs beyond the bench: scientific writing and grant preparation, project and data management, research integrity, and leadership and supervision — the last supported by the EMBO Laboratory Leadership for Postdocs course. I will further participate in the Réseau romand de mentorat pour femmes and the REGARD workshop programme, which support the progression of women towards independent academic positions.

Supervision experience is a deliberate part of the plan: in the second year I will propose and co-supervise a master's student on a defined part of the project, with my supervisor's support, gaining first-hand experience of mentoring and of scoping a project for a student. Together with the teaching experience I bring from Argentina, this prepares me for the training responsibilities of a group leader.

Finally, the fellowship substantially widens my network. Beyond the host group, Switzerland's position and my supervisor's collaborations give me access to the European plant-science community, and I will present my results at international conferences — the International Conference on Arabidopsis Research (ICAR), the EPSO Plant Biology Europe congress, and thermomorphogenesis and plant-development meetings — and at institutional seminars. Combined with my international trajectory across Argentina, France and Switzerland, and a portfolio that will by then span redox biochemistry, developmental imaging and single-cell genomics, this makes me competitive for independent-investigator schemes such as the SNSF Ambizione and Starting Grants and the ERC Starting Grant, whether I continue in Europe or return to Latin America, where these competences are scarce.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Dissemination. The primary route for disseminating results is peer-reviewed publication. I anticipate at least two papers: a first on the characterisation of the temperature response of the SAM (WP1) together with the redox and NO perturbation experiments (WP2), and a second built on the single-nucleus dataset (WP3). Because the project sits at the intersection of two communities, I will target journals read by both plant developmental biologists and the environmental-signalling and redox fields, and I will post preprints at the time of submission so that results reach both communities without delay. I will publish open access and deposit the accepted versions in Libra, the institutional open-access repository of the University of Neuchâtel, in compliance with Horizon Europe requirements.

Results will also be disseminated at the international conferences listed in section 2.1 and at specialised meristem-biology meetings, through seminars at UniNE and partner institutions, and through the collaboration with the Reis group. Presenting to audiences of different composition, from a specialised meristem-biology audience to a broader plant-science one, is itself part of my training in scientific communication.

Research data are a deliverable in their own right. All data underlying publications — imaging data, quantitative phenotyping datasets and the single-nucleus transcriptomic dataset — will be managed under a Data Management Plan following the FAIR principles and deposited in recognised community repositories with persistent identifiers. The single-nucleus dataset of the SAM temperature response (WP3) is designed as a reusable community resource: there are few single-cell datasets of the vegetative shoot apex and none, to my knowledge, addressing temperature, so its value extends well beyond the questions I ask of it here. Protocols and analysis code, including the thermal-time quantification and image-analysis pipelines, will be shared openly so that the methods are reusable.

Exploitation. The project is basic research and no intellectual property is anticipated at the outset. However, leaf-initiation rate is a determinant of canopy establishment and yield, and identifying redox-based

mechanisms that set it could open routes to modulate meristem activity, whether through breeding targets or through chemical modulation of redox signalling. Should an exploitable result emerge, I will assess it with my supervisor and with the research and technology-transfer services of the University of Neuchâtel, and any decision on protection will be taken before publication.

Communication. Communication activities are distinct from dissemination and are directed at audiences beyond the scientific community. I have a sustained record of public engagement — Fascination of Plants Day, university science festivals and open-science advocacy in Argentina — and I will continue this in Neuchâtel through the university's public-engagement programme [specify local formats, e.g. Jardin botanique de Neuchâtel, Fascination of Plants Day, Pint of Science]. The message I will build for general audiences connects the project to something tangible: how plants decide when to make a leaf, and why that decision matters as temperatures rise. I will also contribute to institutional communication channels and press releases where results warrant it, and use social media to reach the wider plant-science and student community. As a woman researcher trained in Latin America and working in Europe, I see value in making that trajectory visible to students considering research careers, and I will seek opportunities to do so.

2.3 The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Scientific impact. The project addresses a gap that is conspicuous once stated: temperature is one of the strongest environmental regulators of plant form, and the shoot apical meristem is where plant form originates, yet the mechanisms linking the two are essentially unknown. Almost everything known about plant thermosensing comes from the hypocotyl, a tissue whose response is driven by cell expansion. By asking the same question in the meristem, where the relevant processes are stem-cell homeostasis, cell division and differentiation, the project tests whether the established thermosensory framework generalises or whether the meristem uses a different logic; my preliminary data point to the latter, which would be a substantive contribution in itself. Beyond this, the project brings redox and nitric oxide signalling into contact with thermomorphogenesis, two fields that have developed largely in parallel, and delivers a single-nucleus dataset of the temperature response of the shoot apex that will serve as a community resource. Because few groups work on the environmental regulation of meristem activity, and because I will work at the interface of the developmental and temperature-signalling communities, I expect the results to be of broad interest across plant biology.

Societal impact. The rate at which leaves are produced determines how quickly a canopy is established, and canopy establishment governs light capture, water use and ultimately yield. This relationship is conserved from *Arabidopsis* to crops such as rice and maize. Under climate change, with rising mean temperatures and more frequent warm episodes, the temperature dependence of leaf initiation becomes directly relevant to how crops perform in the field. Understanding the mechanism that sets this rate is a prerequisite for any strategy that seeks to adjust it, and contributes to the broader European objectives of climate-resilient and sustainable agriculture and of food security. The project also contributes to society through training — it develops a researcher's capacity in an interdisciplinary area — and through outreach that brings plant science to non-specialist audiences.

Economic impact. No direct commercial outcome is expected within the fellowship, as this is fundamental research. In the longer term, knowledge of how meristem activity is set by temperature could inform breeding programmes seeking varieties with architecture optimised for a warming climate or for high-density planting, and the identification of redox-regulated components could provide targets for biotechnological modulation of meristem activity. The fellowship also contributes to the European Research Area by strengthening research capacity at the host institution — installing an *in vivo* redox capability that remains available after the project ends — and by training a researcher whose combined skill set is scarce.

3. Quality and Efficiency of the Implementation

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

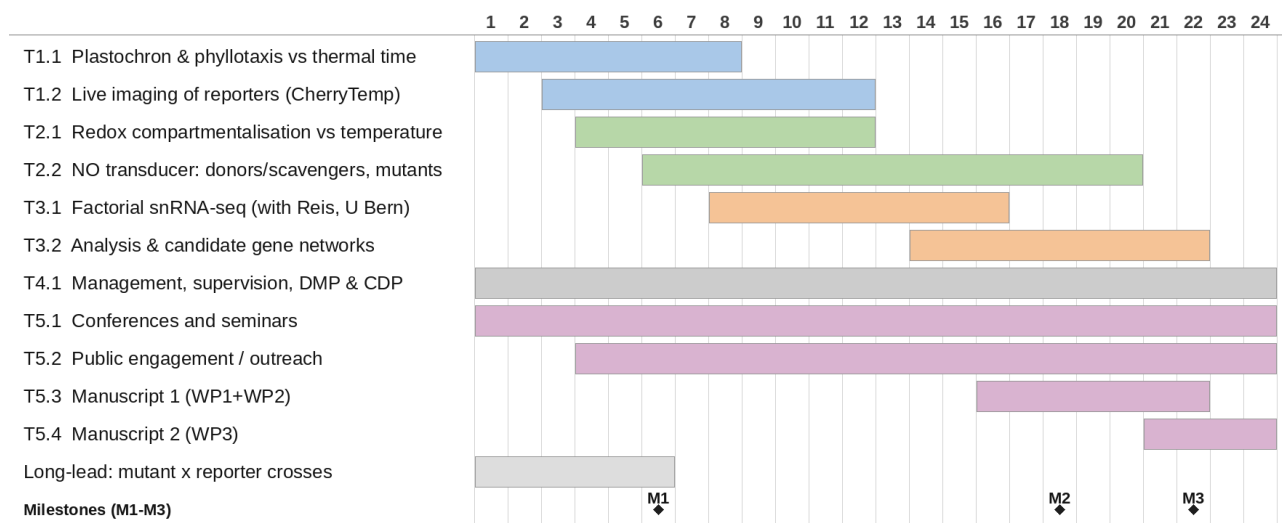
The project is organised into three scientific work packages (WP1–WP3) that together answer one question —

how the shoot apical meristem reads temperature to set the plastochron — plus a management and training work package (WP4) and a dissemination work package (WP5). The scientific WPs are sequenced so that each constrains the next while remaining individually informative. WP1 establishes the temperature response of the meristem on a thermal-time basis and fixes the developmental time windows and reporter lines used downstream. WP2 tests the two candidate transducers — phyB and the redox/NO system — against that baseline. WP3 profiles the response at single-cell resolution under a design that separates temperature from developmental stage. The decisive result of the project — whether NO perturbation reproduces the loss of thermal-time compensation seen in phyB — lies in WP2 and does not depend on the success of the more exploratory WP3.

Interdependencies and critical path. T1.1 (plastochron phenotyping) is the first experiment and the critical-path entry point: it is technically straightforward, uses an assay I have already established, and its per-genotype linear models define which temperatures, time windows and genotypes are carried into WP2 and WP3. Live imaging (T1.2) and the redox/NO experiments (WP2) run in parallel from M3–M6 onward. WP3 tissue collection begins once T1.1 has fixed the two thermal-time points (M8), with a pilot sequencing run before the full factorial. Long-lead work — crosses of the NO-pathway mutants into the pCLV3, pWUS, DR5 and PIN1 reporter backgrounds — starts in M1 to accommodate the 6–8-week Arabidopsis generation time.

Effort. The action supports one researcher for 24 months (24 person-months), distributed approximately as WP1 7 PM, WP2 8 PM, WP3 6 PM, WP4 1 PM and WP5 2 PM. WP2 carries the largest share because it contains the decisive experiment and the widest range of techniques (in vivo imaging, histochemistry, pharmacology, genetics, biotin-switch). WP3 is deliberately bounded: the factorial design and the collaboration with the Reis group (University of Bern) keep the single-cell component to a defined, well-supported task rather than an open-ended screen. The 24-month duration is appropriate — WP1 completes by M12, WP2 by M20, and the WP3 dataset is deposited by M24, leaving the final months for analysis, write-up and the second manuscript.

Table 1. Work-package schedule (months 1–24).



Bars = task active; ♦ = milestone. WP1 blue, WP2 green, WP3 orange, WP4 grey, WP5 pink; long-lead crossing work in light grey. Task titles as in section 1.2.

Table 2. Deliverables (D) and milestones (M).

ID	Title	WP	Month
D1	Career Development Plan	WP4	3
D2	Data Management Plan (FAIR)	WP4	6
D3	Report: temperature response of the SAM (WP1)	WP1	12
D4	Curated dataset: redox/NO maps and perturbation phenotypes	WP2	20
D5	Processed single-nucleus dataset, deposited with a persistent identifier	WP3	24

D6	Manuscript 1 (WP1+WP2) posted as a preprint at submission	WP5	22
D7	Manuscript 2 (WP3) posted as a preprint at submission	WP5	24
M1	Temperatures, time windows and reporter/mutant set fixed for WP2–WP3	WP1	6
M2	Decisive experiment concluded: does NO perturbation phenocopy phyB?	WP2	18
M3	Temperature- and thermal-time-responsive gene networks of the SAM defined	WP3	22

Table 3. Risk assessment (L = low, M = medium, H = high).

Risk	L	I	Mitigation
The phyB thermal-time phenotype does not hold under controlled temperature shifts, or effect sizes are small.	L	M	The phenotyping assay is already established and gives robust per-genotype linear models; multiple constant temperatures and shift regimes; increased replication; independent readouts (leaf number, SAM size, reporter domains).
NO perturbation produces no plastochron phenotype.	M	M	A null result still discriminates between models and is publishable: Task 2.1 independently tests the ROS branch, and pharmacology, genetics and reporter localisation triangulate. The glutathione control separates NO-specific from general thiol-redox effects.
Single-nucleus RNA-seq under-samples the meristem, or nuclei yield from the vegetative apex is low.	M	M	Micro-dissection enrichment of the apex; pilot run at M8 before the full factorial; the Reis group has aerial-tissue protocols; fallback to reporter-sorted low-input RNA-seq of zonal populations.
Delay or reduced availability in the Reis collaboration.	L	M	Written work plan and scheduled technical visits from M1; sequencing can be outsourced to a core facility; analysis pipelines are portable and partly covered by SIB courses.
CherryTemp live-imaging of the apex is technically demanding (drift, phototoxicity, z-range).	M	L	The system and confocal are installed and in routine use in the host lab; start with stable set-points before rapid shifts; WP1's core result (T1.1) does not depend on live imaging.
Arabidopsis line generation slower than the 6–8-week cycle allows.	M	L	Reporter and mutant lines are already in the host lab; crosses started in M1; staggered sowing.
24 months too short for two manuscripts.	L	M	Manuscript 1 (WP1+WP2) is self-contained and targeted for M22; the WP3 dataset is a standalone deliverable (M24) whose paper may be completed shortly after tenure.

3.2 Quality and capacity of the host institution and participating organisations, including hosting arrangements

Host institution. The fellowship is hosted by the group of Prof. Martina Legris at the Institute of Biology, Faculty of Sciences, University of Neuchâtel (UniNE). The institute is an established centre for plant biology in Switzerland, with several groups working on plant development, physiology and plant–environment interactions, a shared seminar programme, and membership of the CUSO doctoral programme in plant sciences and the Swiss Plant Science Web. This gives the fellow an immediate community in plant developmental and environmental biology and structured doctoral- and postdoctoral-level training.

Research environment and equipment. The host group is funded by an SNSF Starting Grant (“Environmental control of shoot architecture in Arabidopsis”, 2023–2028) and a two-year Fondation Mercier pour la Science grant that establishes the temperature-and-leaf-initiation line this fellowship contributes to. The infrastructure the project needs is in place: a spectral confocal microscope co-funded by the supervisor

(SNSF R’equip, 2024); a CherryTemp sample-level temperature-control system for live imaging; controlled-environment growth chambers with temperature logging; and full molecular-biology, histochemistry and plant-transformation facilities. The reporter lines (pCLV3, pWUS, DR5, PIN1) and the thermosensor and NO-pathway mutants required by the project are already available in the group.

Supervision. Prof. Legris is a foundational contributor to plant thermomorphogenesis — first author of the work establishing phyB as a thermosensor⁴ and of subsequent work on light and temperature control of leaf and meristem morphogenesis^{5–6} — and holds the 2023 *New Phytologist* Tansley Medal. As a former MSCA, EMBO and HFSP fellow she knows the instrument from the inside. The group comprises the supervisor, one postdoctoral researcher and one PhD student; supervision is organised around weekly one-to-one and group meetings, a Career Development Plan reviewed at regular intervals, and quarterly progress presentations, with co-supervision of a master’s student planned for the fellow in year 2 (see 1.3).

Career and open-science support. The UniNE Graduate Campus (swissuniversities 2025–2028 programme) provides transferable-skills and career training; the university’s research and technology-transfer services support grant preparation and any exploitation questions; and the Libra institutional repository ensures Horizon Europe-compliant open access. The fellow will also access CUSO and UniNE doctoral-programme courses and the Réseau romand de mentorat pour femmes and REGARD programmes.

Participating organisation — University of Bern (collaboration). The single-nucleus transcriptomics of WP3 is carried out with the group of Dr R. Reis at the University of Bern (~40 minutes from Neuchâtel), which provides single-cell/nucleus methodology and computational analysis. This is a scientific collaboration involving short technical visits, not a beneficiary or associated-partner role, and no secondment is foreseen.

References

- 1 Wenzl and Lohmann (2023), *Cells & Development*. 175:203850
- 2 Zeng et al. (2017), *EMBO Journal*. 36:2844–2855
- 3 Zeng et al. (2023), *Nature Communications*. 14:8001
- 4 Legris et al. (2016), *Science*. 354:897–900
- 5 Legris et al. (2019), *Nature Communications*. 10:5219
- 6 Legris (2023), *New Phytologist*. 240:2191–2196
- 7 Iglesias et al. (2024), *PNAS*. 121:e2320187121
- 8 Luquet et al. (2025), *Plant Science*. 352:112377
- 9 Nejamkin et al. (2025), *Journal of Plant Growth Regulation*. 44:5147–5159

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Part B-2

(No overall page limit applied)

4. CV of the researcher

Insert here text for your proposal

5. Capacity of the Participating Organisation(s)

5.1 Template table: Overview of Participating Organisations

Organisation role	PIC	Legal Entity Short Name	Academic organisation (Y/N)	Country	Name of Supervisor
Beneficiary					
Associated partner linked to a					

beneficiary (if applicable)					
Associated partner for outgoing phase (mandatory for GF)					
Associated partner for secondment (if applicable)					
Associated partner for non-academic placement (if applicable)					

5.2 Template table: Capacity of the Participating Organisations

Maximum 1 page per Beneficiary and ½ page per Associated partner.

Choose one of: ? Beneficiary (compulsory) ? Associated partner linked to a beneficiary (if applicable) ? Associated partner for outgoing phase (mandatory for GF only) ? Associated partner for secondment (if applicable) ? Associated partner for non-academic placement (if applicable)	
[Full name + Legal Entity Short Name + Country]	
General description	
Role and profile of supervisor	
Key research facilities, Infrastructure and Equipment	
Previous and current involvement in EU-funded research and training programmes/actions/projects	

6. Additional ethics information

Insert here text for your proposal

(NB: Only if you have additional information that could not be included in the ethics self-assessment)

7. Additional information on security screening

Insert here text for your proposal

(NB: Only if you answered yes to one of the questions in the security issues table, with the exception of “Does this activity involved HE associated and/or third countries?”)

8. Environmental considerations in light of the MSCA Green Charter

Insert here text for your proposal

9. Required for Global Fellowships only: Letter of commitment from the associated partner hosting the outgoing phase

Insert here the Letter of commitment (if applicable)